

Application for patent filing

Date	D	D	M	M	Y	Y	Y	Y
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Name of the Faculty	: Dr. R. Siva
Department	: Computer Science and Intelligent Systems (CINTEL)
Faculty ID Number	: [Faculty ID — to be filled by Dr. R. Siva]
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Contact no. of all Inventors	Adnan Nagdiwala: [contact no.] Chandra Bhaya: [contact no.]
Major area of invention	: Precision Agriculture / Agricultural Technology (AgriT)
Narrow focus area of invention	: IoT-based soil sensing, ML crop recommendation & r
Title of the invention	CropXpert: An IoT-Enabled, ML-Driven Precision Agr
Earlier status of research	: Prior crop recommendation systems (Pudumalar et a
How different your invention from similar research / others - Novelty?	: CropXpert is the first system to simultaneously: (1) in
Possible domain for field application	: Agriculture, Rural Technology, Smart Farming, IoT, A
Possible sector for commercialization	: AgriTech startups, State Agriculture Departments, N/
Faculty Signature with date	: Dr. R. Siva [Signature] Date: 21 / 04 / 202

Invention Disclosure Form

To be filled by the inventors

Please provide highly relevant information for details asked below and use consistent language while describing the specific feature or element in the invention disclosure.

1. **Title of invention** (Please indicate a title for the invention and technology of the invention)

CropXpert: An IoT-Enabled, Machine Learning-Driven Precision Agriculture Platform with

2. **Describe the invention.** (Please describe specifically about the general purpose of invention. Is the invention a new process, device or product, system, software or a combination of these elements?)

CropXpert is an integrated hardware-software system combining: (i) an IoT sensor device

3. Does the invention provide a **new use of or improvement to an existing product or process**? (Highlight the use or improvements from the existing with recent and relevant references)

Yes — CropXpert improves upon existing crop recommendation systems in three key ways

4. State the **Novelty** of the invention and specify the claims in the invention

NOVELTY: First integrated platform to combine IoT soil sensing + ML crop advisory + live

5. Describe the **advantages of the present invention over the existing technologies** (please identify the advantages e.g. efficiency, cost benefits, simplicity etc.

Cost: Complete IoT unit ~INR 3,600 vs INR 500-2,000 per parameter for lab tests (reuse)

6. Describe how the **present invention overcomes the drawbacks** of currently available technology related to your invention. (please include the relevant references)

Drawback 1 (Market blindness): Prior systems [Doshi 2018; Pudumalar 2017] recommend

7. Describe **uses, applications and benefits** of the invention.

SRM Institute of Science & Technology
Application of CropXpert as a crop recommendation advisory for 100M+ Indian smallholders; real-time soil health

8. Does the focus of the invention results in **societal impact technology**? (Please describe how in detail, also specify the commercial applications, market need of product/ service of invention and why?)

Yes. CropXpert directly addresses SDG 2 (Zero Hunger) by improving crop yields for 14

9. Characterize the **disadvantages and limitations** of the invention

1. Hardware requires initial INR 3,600 investment — mitigated by PM-KISAN subsidy at

10. Enclose the **sketches, drawings, photographs** and other materials that help in better understating/ illustration of the novelty in the invention.

Enclosed (as separate annexure): Fig 1 — System Architecture Diagram (5-layer Micro:

11 **Current development status of the invention**

- A. Has your invention been tested experimentally

Yes. Sprint 1 (Sensor Integration) fully completed and validated: all 5 stories accepted.

- B. Describe the experimental approach of the invention also state the methods adopted in the experiment.

Hardware: Bench-top testing of RS485/Modbus NPK sensor with MAX485 converter and

- C. Are the experimental data is documented in a formal log or any instrumental confirmation available for the invention (kindly provide the details)

Yes. Formal documentation maintained in: (1) Sprint Retrospective Log (Sprint 1) — rec

- D. Is further development of your invention is necessary or development of the invention is in progress (provide the relevant information)

Yes — development is in progress (Sprint 2 active). Planned: ESP32 Wi-Fi reconnect h

12. **Please list any of your publications** (including abstracts, posters, news releases, etc.) to emphasize the present invention background.

[1] A. Nagdiwala and C. Bhayal, "CropXpert: An IoT-Enabled, Machine Learning-Driven

13. INVENTOR(S) AND/OR CONTRIBUTOR(S):

	INVENTOR (1)	INVENTOR (2)
Signature:		
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Citizenship	INDIAN	INDIAN
(Country):		

14. ASSIGNMENT DETAILS: Assignee is the entity or individual who holds the patent.

Signature: (To be signed by the authorized signatory on behalf of the assignee)

Name of the Authorized Signatory and Designation: Director / Dean, School of Computing, SRMIST Katta

Address: SRM Institute of Science and Technology, Kattankulathur

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